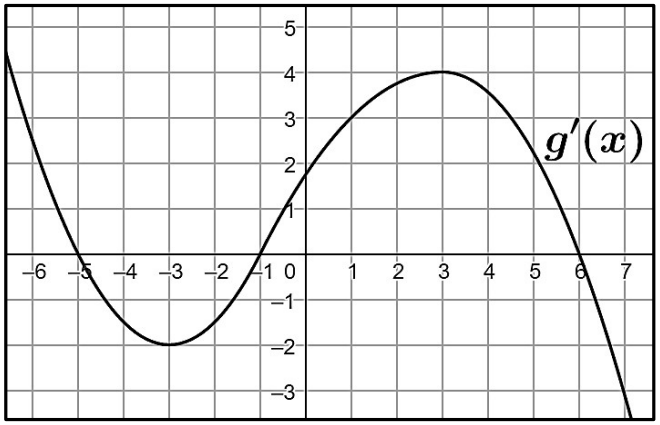
Worksheet 5-6 B Concavity and Points of Inflection #2
AP Calculus

Name SOLUTIONS



1. The graph of $g'(x)$, the derivative of g, is continuous and differentiable for all real numbers and is given above. $g'(x)$ has horizontal tangents at $x = -3$ and $x = 3$.

(A) On what interval(s) is $g(x)$ decreasing? Give a reason for your answer.

$g(x)$ is decreasing on $[-5, -1]$ and $[6, +\infty)$ because $g'(x) \leq 0$ on these intervals

(B) At which value(s) of x does $g(x)$ have a relative maximum? Explain your reasoning.

$g(x)$ has a relative maximum at $x = -5$ and $x = 6$ because $g'(x)$ changes from pos to neg at these x-values.

(C) On what interval(s) is $g(x)$ concave up? Give a reason for your answer.

$g(x)$ is concave up on $(-3, 3)$ because $g'(x)$ is increasing on this interval

(D) On what interval(s) is $g(x)$ concave down? Explain your reasoning.

$g(x)$ is concave down on $(-\infty, -3)$ and $(3, +\infty)$ because $g'(x)$ is decreasing on these intervals

(E) At which value(s) of x does $g(x)$ have a point of inflection? Explain your reasoning.

$g(x)$ has a point of inflection at $x = -3$ because $g'(x)$ changes from decreasing to increasing, and at $x = 3$ because $g'(x)$ changes from increasing to decreasing.

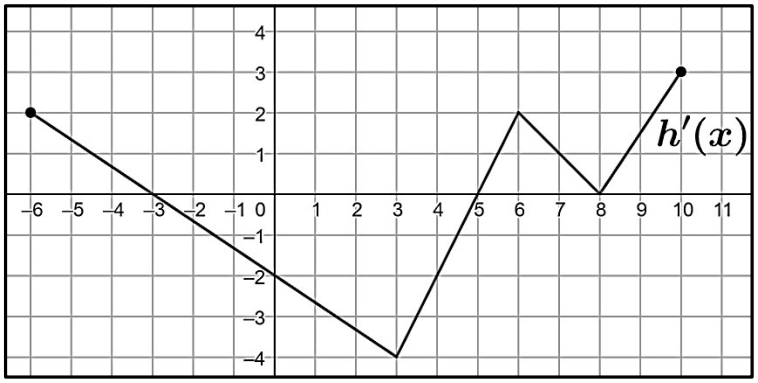
(F) On what interval(s) is $g(x)$ both increasing and concave down? Give a reason for your answer.

$g(x)$ is increasing and concave down on $(-\infty, -5)$ and $(3, 6)$ because $g'(x) > 0$ and decreasing on these intervals.

(G) On what interval(s) is $g(x)$ both decreasing and concave up? Give a reason for your answer.

$g(x)$ is decreasing and concave up on $(-3, -1)$ because $g'(x) < 0$ and increasing on these intervals.

Created by B Passwater edited by K Bakowski



2. The graph of $h'(x)$, the derivative of h, consists of four line segments and is given above.

(A) On what interval(s) is $h(x)$ increasing? Give a reason for your answer.

$h(x)$ is increasing on $[-6, -3]$ and $[5, 10]$ because $h'(x) \geq 0$ on these intervals

(B) On what interval(s) is $h(x)$ decreasing? Give a reason for your answer.

$h(x)$ is decreasing on $[-3, 5]$ because $h'(x) \leq 0$ on this interval

(C) At which value(s) of x does $h(x)$ have a relative minimum? Explain your reasoning.

$h(x)$ has a relative minimum at $x = 5$ because $h'(x)$ changes from neg to pos at this x-value.

(D) At which value(s) of x does $h(x)$ have a relative maximum? Explain your reasoning.

$h(x)$ has a relative maximum at $x = -3$ because $h'(x)$ changes from pos to neg at this x-value.

(E) On what interval(s) is $h(x)$ concave up? Give a reason for your answer.

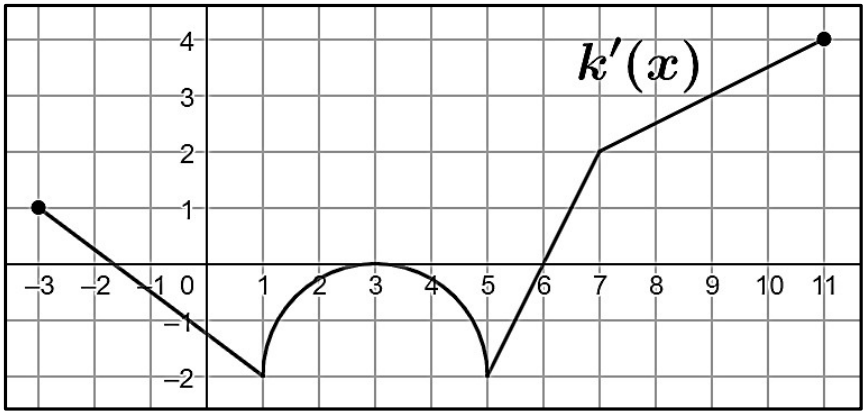
$h(x)$ is concave up on $(3, 6)$ and $(8, 10)$ because $h'(x)$ is increasing on these intervals

(F) On what interval(s) is $h(x)$ concave down? Explain your reasoning.

$h(x)$ is concave down on $(-6,\ 3)$ and $(6,\ 8)$ because $g'(x)$ is decreasing on these intervals

(G) At which value(s) of x does $h(x)$ have a point of inflection? Explain your reasoning.

$h(x)$ has a point of inflection at $x = 3$ and $x = 8$ because $h'(x)$ changes from decreasing to increasing, and at $x = 6$ because $h'(x)$ changes from increasing to decreasing.



3. The graph of $k'(x)$, shown above, consists of three line segments and a semi circle.

(A) At which value(s) of x does $k(x)$ have a point of inflection? Explain your reasoning.

$k(x)$ has a point of inflection at $x = 1$ and $x = 5$ because $k'(x)$ changes from decreasing to increasing, and at $x = 3$ because $k'(x)$ changes from increasing to decreasing.

(B) Order the following values from least to greatest: $k''(3)$, $k''(4)$, $k''(6)$.

$k''(3) = 0,\ k''(4) < 0,\ k''(6) > 0 \ \rightarrow\ k''(4) < k''(3) < k''(6)$

(C) Order the following values from least to greatest: $k''(-2)$, $k''(6)$, $k''(9)$.

$k''(-2) < 0,\ k''(6) > 0,\ k''(9) > 0 \ \rightarrow\ k''(-2) < k''(9) < k''(6)$

steeper so bigger positive

(D) Order the following values from least to greatest: $k'(-2)$, $k'(-1)$, $k'(3)$.

$k'(-2) \approx 0.3,\ k'(-1) = -0.5,\ k'(3) = 0$

$\rightarrow\ k'(-1) < k'(3) < k'(-2)$

(E) Order the following values from least to greatest: $k'(0)$, $k''(2)$, $k'(6)$.

$k'(0) \approx -1.2,\ k''(2) > 0,\ k'(6) = 0$

$\rightarrow\ k'(0) < k'(6) < k''(2)$

(F) For each of $k'(5)$ and $k''(5)$, find the value or explain why it does not exist.

$k'(5) = -2$,

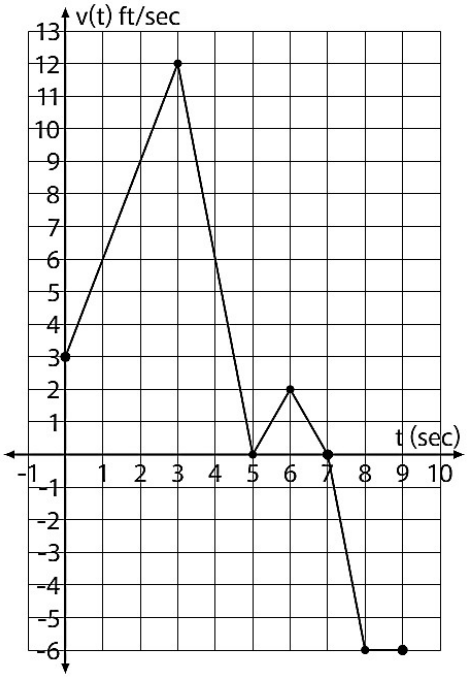
$k''(5)$ does not exist because the graph of $k'(x)$ has a corner or cusp

(G) For each of $k'(0)$ and $k''(0)$, find the value or explain why it does not exist.

$k'(5) = -2$,

$k''(5)$ does not exist because the graph of $k'(x)$ has a corner or cusp

4. The graph above represents the velocity v, in feet per second, of a particle moving along the x-axis over the time interval for $0 \le t \le 9$ seconds.



(A) At $t = 4$ seconds, is the particle moving to the right or left? Justify.

$v(4) = 6$, the particle is moving to the right because $v(6) > 0$.

(B) At what time(s) is the particle at rest? Justify.

the particle is at rest at $t = 5$ and $t = 7$ because $v(t) = 0$ at these t-values

(C) At what time(s) does the particle change direction? Justify.

the particle changes direction at $t = 7$ because $v(t)$ changes from positive to negative at this t-value

(D) On what open intervals $0 < t < 9$ is the particle moving left? Justify.

the particle is moving left on (7, 9) because $v(t) < 0$ on this interval

(E) What is the acceleration of the particle at $t = 4$ seconds? Show the work that leads to your answer.

$a(4) = v'(4) = \frac{0-12}{5-3} = -\frac{12}{2} = -6$ ft/s^2

(F) On what open intervals $0 < t < 9$ is the acceleration of the particle positive? Justify.

$a(t) = v'(t)$ is positive on $(0, 3)$ and $(5, 6)$ because $v(t)$ is increasing on these intervals

(G) What is the **average acceleration** of the particle over the interval [3,6] seconds? Show the computations that lead to your answer and indicate units of measure.

Average acceleration = Average rate of change of velocity

$= \frac{v(6)-v(3)}{6-3} = \frac{2-12}{6-3} = \frac{-10}{3}$ ft/s^2

(H) On what open intervals $0 < t < 9$ is the **speed** of the particle decreasing? Justify.

Speed is decreasing when $v(t)$ and $a(t)$ have opposite signs

The speed of the particle is decreasing on $(3, 5)$ and $(6, 7)$ because $v(t) > 0$ and $v(t)$ is decreasing (meaning $a(t) = v'(t) < 0$).